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New Spectroscopic Analyses of the $A^1\Sigma^+$–$X^1\Sigma^+$ Bands of ^{7}LiH[1]

F. B. ORTH AND W. C. STWALLEY

Departments of Chemistry and Physics, University of Iowa, Iowa City, Iowa 52242

The emission spectrum of ^{7}LiH has been photographed in the ~5200- to 8700-Å region and the previously taken spectrum in the ~3400- to 5300-Å region has been re-examined. Bands of the $A^1\Sigma^+$–$X^1\Sigma^+$ electronic transition have been analyzed and the observed vibrational structure of the ground state has been extended from $v'' = 5$ to $v'' = 12$. Rotational levels significantly higher than those previously reported have been observed, and the higher rotational levels of $v' = 0$, 1, and 2 of the $A^1\Sigma^+$ state were found to be anomalous. This anomaly is attributable to a breakdown of the rotational expansion for the anomalous $A^1\Sigma^+$ potential. A comparison of observed and theoretical relative band intensities reveals discrepancies presumably related to rotational shifts in transition probabilities, inaccuracies in the vibrational wavefunctions for low v' of the $A^1\Sigma^+$ state, and/or inaccuracies in the A–X electronic transition moment. New RKR turning points have been calculated up to $v'' = 12$ of the $X^1\Sigma^+$ state. A cumulative list of all lines that have been identified belonging to the $A^1\Sigma^+$–$X^1\Sigma^+$ transition of ^{7}LiH is included.

I. INTRODUCTION

The emission spectrum of ^{7}LiH has been photographed in the ~5200- to 8700-Å region using the high resolution obtainable from a 3.4-m Ebert spectrograph, and bands of the $A^1\Sigma^+ \rightarrow X^1\Sigma^+$ electronic transition have been analyzed. The previous spectroscopic analysis of the A–X system of ^{7}LiH consists of the following:

1. Crawford and Jorgensen (*1*) photographed ^{7}LiH (and also ^{7}LiD) in absorption and emission in the ~3200- to 4300-Å region. They reported data on 26 bands that covered v'' from 0 to 3 and v' from 0 to 14 (additionally, lines belonging to the (15–0) band were reported in Jorgensen's thesis). Their work was a reinvestigation of lower-dispersion work done previously by Nakamura (*2*, *3*).

2. Li and Stwalley (*4*) photographed ^{7}LiH (and also ^{7}LiD, ^{6}LiH, ^{6}LiD) in emission in the ~3400- to 5300-Å region. They analyzed only the ~4700- to 5300-Å region and identified 7 new bands extending the observed ground-state vibrational structure from $v'' = 3$ to $v'' = 5$. They also refitted lines identified by Crawford and Jorgensen using a "direct approach" analysis rather than the conventional methods of rotational analysis. The rotational constants and band origins that they obtained agreed completely with those previously reported by Crawford and Jorgensen.

The previously unanalyzed spectrograms by Li and Stwalley from ~3400

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to 4700 Å have also now been analyzed for $A-X$ bands, primarily for the following purposes: (i) to fill the ~4300- to 4700-Å gap and thus obtain more information on $v'' = 3$, and (ii) to extend as much as possible the rotational structure of previously identified branches, and (iii) to make an intensity analysis of various bands in this region.

In all, 32 new bands have been identified, and the observed vibrational structure for the ground state has been increased from $v'' = 5$ to $v'' = 12$. Experimental data on $v' = 0$ of the A state and on $v'' = 2, 3, 4$, and 5 of the X state have been significantly increased (especially $v' = 0$ and $v'' = 3$). Major rotational extensions have been made on 9 previously identified bands and another 8 new but fragmentary bands have been found.

In the Appendix all lines, arranged by bands, that have been identified as belonging to the $A^1\Sigma^+-X^1\Sigma^+$ transition of ^{7}LiH are recorded.

II. EXPERIMENTAL DETAILS

The experimental procedure used to obtain the spectrum of ^{7}LiH in the ~5200- to 8700-Å region is similar to that previously described (*4*). Kodak 103 a-D, 103 a-F, and hypersensitized 1-N spectroscopic plates were used for detection, and neon and thorium lines emitted from a Westinghouse thorium hollow cathode lamp were used for standards. The spectrograms were taken in the fourth, fifth, and sixth orders of a 3.4-m Ebert spectrograph with reciprocal dispersions of ~2.2, 1.7, and 1.4 Å/mm, respectively. Commercial LiH containing the natural proportion of ^{6}Li to ^{7}Li and ~30 Torr of H_2 were placed in the discharge tube. Exposure times were limited by the duration of vaporization of the LiH sample, usually 3 to 4 hr. Considerable difficulty was experienced in getting satisfactory spectrograms, and many runs were failures. There is a significant difference in the quality and strength of the various spectrograms which were finally chosen to be measured; the spectrograms in the ~6850- to 7250-Å region are quite poor. A considerable problem also exists in the neighborhood of the Li resonance line at 6707 Å. The best spectrogram obtained in this neighborhood still has 70 Å of structure completely obscured by the wings of this atomic line.

The lines in the ~5200- to 8700-Å region were measured at Argonne National Laboratory using a manually operated plate comparator equipped with a photoelectric setting device (*5*). Such a comparator can reproduce line settings to 0.5 μm. The agreement in absolute position of sharp lines in overlapping regions of different spectrograms averages ~0.03 cm^{-1}. From 85 to 90% of the stronger lines in the ~5200- to 7000-Å region have been identified; however, in the 7000- to 8700-Å region many of the stronger lines are still unidentified. No evidence of the spectrum of ^{6}LiH has been observed. A listing of all measured lines (also showing the assignment for each of the identified lines) may be obtained on request from these authors.

III. INTENSITY ANALYSIS

The lines in the ~3400- to 5300-Å region photographed by Li and Stwalley were measured using an automatic scanning comparator with an encoder system and on-line computer service at the Argonne National Laboratory. The

TABLE I

Relative Intensities of Selected Bands in the $A^1\Sigma^+$–$X^1\Sigma^+$ System of ^{7}LiH

v' \ v''	1	2	3	4	5	6	7	8	9	10	11	12
0	a	a	33	167	268	500[b]	700[b]	700[b]	500[b]	300[b]	150[b]	a
	40	118	249	417	573	679	700	628	491	334	197	99
1	23	97	325	563	580	800[b]	500[b]	200[b]	a	c	100[b]	40[b]
	203	465	711	784	613	314	68	3	113	273	356	324
2	151	252	557	473	468	150[b]	40[b]	400[b]	200[b]	c	a	a
	513	833	791	405	53	38	256	377	254	55	9	136
3	451	379	504	148	40	40[b]	150[b]	300[b]	a	c		
	844	857	346	3	179	420	295	36	46	229	235	64
4	535	431	272	12	181	200[b]	a	a	100[b]	60[b]		
	1000	497	7	235	464	196	2	196	277	73	18	178
5	326	171	46	88	148	20[b]		100[b]				
	888	104	163	503	193	15	277	234	6	123	212	37
6	148	20	35	157		100[b]						
	568	12	483	306	4	298	228	0	193	176	0	136
7	47					200[b]						
	222	232	510	21	246	291	1	212	169	4	179	85

[a] Could not be found, presumably due to insufficient intensity.

[b] Eye estimate.

[c] Branches fell in poor plate region making reasonable estimate not possible.

Note. The first entry for each band is the observed relative intensity. The second entry is the theoretically calculated value of $\nu_{nm}^4 |D_{nm}|^2$ scaled to a value of 700 for the (0–7) band.

output from this device includes numerical values of line intensities. By integrating the intensities of lines belonging to a given band some observed relative band intensities of A–X bands to the blue of 5300 Å were obtained, and they are given in Table I.

Only positions for lines in the ~5200- to 8700-Å region were obtained using the comparator discussed in Section II, so that in this region only eye-estimates of observed relative band intensities were possible. These eye-estimates are also given in Table I (Attempts to use the automatic scanning comparator in this region failed to produce consistent results. This failure may have been caused by the heavy and uneven background in many of these plates due, for example, to hypersensitizing of plates, continuum from the Li atomic resonance line, etc.).

The intensity of a band is a summation of line intensities. The intensity of an emission line has the proportionality

$$I_{nm} \sim N_n \nu_{nm}^4 |D_{nm}|^2,$$

where N_n is the number of molecules in the initial state, ν_{nm} is the frequency of transition, and D_{nm} is the transition dipole matrix element. The quantities $\nu_{nm}^4 |D_{nm}|^2$ are molecular constants which have been calculated (*6*) theoretically for all $A-X$ bands of ^{7}LiH, and these quantities, scaled to a value of 700 for the 0–7 band for convenient comparison with observed quantities, are also given in Table I (note they do *not* include the factor N_n).

The observed and theoretical quantities discussed above can be easily compared along any row of a Deslandres table since in emission N_n is a constant within a row. It can thus be seen from Table I that the agreement between the observed and theoretical quantities would be significantly improved if all theoretical quantities were shifted over one column (vibrational quantum) to the right. This is particularly evident from the positions of the intensity minima in the v' = 1, 2, 3, and 4 rows shown in Table I. It is also true for the v' = 0 row since, in this progression, the two largest observed values occur at the (0–7) and (0–8) bands, while the two largest theoretical values occur at the (0–6) and (0–7) bands. Two other discrepancies are noted in Table I. A comparison of the observed and theoretical quantities in the v' = 0 row suggests that the actual vibrational wavefunction for v' = 0 of the $A^1\Sigma^+$ state is narrower and steeper than the theoretical wavefunction. Also, the observed (1–12) band intensity is considerably weaker than that predicted, whether or not the theoretical values are shifted by one column as mentioned above.

It should be realized that the observed relative band intensities in Table I cannot be taken too quantitatively since their determination is hindered by blending, obscuring, background differences, plate sensitivity differences (although overlap from plate to plate greatly facilitates comparison of different plates), etc. Nevertheless, the description, in Table I, of the positions of the very strong and very weak observed bands is quite unambiguous. Moreover, for those bands whose relative intensities were determined from integration of numerical output of line intensities, it is hard to accept that the ratios of consecutive intensities within a row would be off by more than 25%. Thus it is felt that the above discrepancies indicate some problems with the reported theoretical values.

The discrepancies in Table I are felt to be the result of an unknown combination of three factors: (i) variation of transition probability with J'; (ii) inaccuracies in the potential minimum region of the $A^1\Sigma^+$ potential; and (iii) inaccuracies in the ab initio electronic transition moment function.

We have compared unpublished theoretical transition probability calculations for J' = 11 to J'' = 10 by Warren T. Zemke in our laboratory with those given in Table I (for J' = 0 to J'' = 0). The rotational shifts do indeed improve agreement with experiment (e.g., (3–4) and (3–5) bands are predicted to be comparable in intensity) but do not completely resolve the discrepancies (e.g., the band (2–7) is still predicted to be much more intense than the band (2–6), whereas the opposite is observed experimentally). Thus attempts are being made to construct an improved $A^1\Sigma^+$ potential. It is felt that the $A-X$ bands are a very good system for future study, both theoretical and experimental, of absolute and relative radiative transition probabilities.

IV. MOLECULAR CONSTANTS

Band data were fit by least squares to the following truncated power series:

$$\frac{R(J) - P(J)}{J + \tfrac{1}{2}} = \frac{\Delta_2 F'(J)}{J + \tfrac{1}{2}} = \sum_{i=0}^{2} a_i'(J + \tfrac{1}{2})^{2i}, \tag{1}$$

$$\frac{R(J-1) - P(J+1)}{J + \tfrac{1}{2}} = \frac{\Delta_2 F''(J)}{J + \tfrac{1}{2}} = \sum_{i=0}^{2} a_i''(J + \tfrac{1}{2})^{2i}, \tag{2}$$

$$R(J) + P(J) = \sum_{i=0}^{3} b_i[J(J+1)]^i. \tag{3}$$

The coefficients of these power series have the following relationships to molecular constants (*7*),

$$a_0 = 4B_v - 6D_v + {}^{27}/_4 H_v,$$

$$a_1 = -8D_v + 34H_v,$$

$$a_2 = 12H_v,$$

$$b_0 = 2(\nu_0 + B_v' - 2D_v' + 4H_v'),$$

$$b_1 = 2(B_v' - B_v'' - 6D_v' + 18H_v'),$$

$$b_2 = 2(D_v'' - D_v' + 15H_v'),$$

$$b_3 = 2(H_v' - H_v''),$$

where ν_0 is the origin of a band and B_v, D_v, and H_v are the truncated-power-series coefficients (effective rotational constants) for the rotational energy of vibrating rotator, $F_v(J)$:

$$F_v(J) = B_v J(J+1) - D_v J^2(J+1)^2 + H_v J^3(J+1)^3. \tag{4}$$

Thus the lower-state rotational constants were first determined from Eq. (2). The upper-state rotational constants determined from Eq. (1) were substituted into Eq. (3) in order to obtain ν_0 and second values of the lower-state rotational constants. Since band data on $v'' = 2$, 3, 4, and 5 of the X state have been increased significantly it was decided to determine molecular constants related to these levels even though these constants have been previously reported (*4*). The experimental constants related to $v'' = 6$ to 12 are new.

The band origins are given in Table II as the upper entry for each band. The lower entry for each band gives the value of J_{max} and twice the standard deviation in the fitting to Eq. (3), respectively. J_{max} is *not* the maximum value of J observed for a given band (except in cases where high rotational lines could not be found), but rather, J_{max} is the maximum value of J in the subset of band data used to obtain the band origin and rotational constants. It was found that if band members corresponding to values of J larger than a certain value were included in the least-squares fit, unacceptably large standard deviations resulted which increased exponentially with increasing J_{max}. On the basis of the details of this problem, which will be discussed further in Section V, and the known fact that in general the values of molecular constants do depend slightly on the

TABLE II

Band Origins of the $A^1\Sigma^+ - X^1\Sigma^+$ System of ^7LiH

v'' / v'	2	3	4	5	6	7	8	9
0		21997.81 16, 0.08	20770.04 17, 0.15	19584.59 17, 0.10	18440.69 16, 0.08	17338.10 17, 0.12	16276.39 16, 0.10	15255.23 17, 0.08
		1227.77	1185.45	1143.90	1102.59	1061.71	1021.16	980.24
		280.72	280.81	280.75	280.91	280.93	280.87	
1	23549.46 17, 0.09	22278.53 18, 0.19	21050.85 17, 0.11	19865.34 17, 0.09	18721.60 17, 0.05	17619.03 18, 0.12	16557.26 18, 0.13	
	1270.93	1227.68	1185.51	1143.74	1102.57	1061.77		
	312.98	313.04	312.85	312.96			313.00	
2	23862.44 17, 0.04	22591.57 17, 0.10	21363.70 17, 0.12	20178.30 16, 0.08			16870.26 17, 0.06	15849.10 16, 0.10
	1270.87	1227.87	1185.40				1021.16	
	335.69	335.69	335.62	335.91			335.55	
3	24198.13 17,0.04	22927.26 13, 0.11	21699.32 17, 0.10	20514.21 13, 0.18	19370.57 11, 0.14	18267.84 17, 0.08	17205.81 17, 0.12	
	1270.87	1227.94	1185.11[a]	1143.64[a]	1102.73	1062.03[a]		
	352.82	352.90		352.73	352.52			
4	24550.95 17, 0.04	23280.16 17, 0.08		20866.94 16, 0.14	19723.09 16, 0.11			16537.59 16, 0.10
	1270.79			1143.85				980.59[b]
	366.14			365.70				
5	24917.09 18, 0.27		22417.88 14, 0.12	21232.64 17, 0.16			17924.83 16, 0.13	
			1185.24					
			375.72					
6			22793.60 17, 0.09		20464.45 13, 0.18			
					382.94			
7					20847.39 17, 0.18			

v'' / v'	10[b]	11	12
0	980.24 14274.99 17, 0.07	939.97 13335.02 17, 0.09	
		280.86	
1		13615.88 17, 0.05	897.97 12717.91 10, 0.14

[a] These values were not used in the averaging due to the poor quality of associated bands.

[b] The 4-10 band origin (not shown) is 15557.00, J_{max} is 17, and 2σ is 0.15.

Note. The upper entry for each band is the band origin (in cm^{-1}). The lower entries are the values of J_{max} (used to obtain the molecular constants from band data) and two standard deviations (in cm^{-1}) from the fit to Eq. (3). Underlined numbers between bands are $\Delta G'$ or $\Delta G''$ values.

highest value of J in the fitting, it was decided to use J_{max} values of about 17 as consistently as possible.

Of the band origins in Table II that have been previously reported (*4*), the agreement of values is most often less than 0.03 cm^{-1}. The important exceptions to this are the (2–4) and (3–4) origins for which the disagreements are 0.41 and 1.67 cm^{-1}, respectively. The (2–4) and (3–4) band origins shown in Table II are the correct values because they remove the significant inconsistency in the ΔG values shown in Table VII of Ref. (*4*). The $\Delta G'$ and $\Delta G''$ values are given here as underlined numbers in Table II. Averages of these ΔG values are given in Table III.

The rotational constants B_v determined from fits to Eqs. (1)–(3), are given, respectively, for each band in Table IV. The values of J_{max} used to obtain these values correspond to the J_{max} values listed in Table II. Similarly, the rotational constants D_v and H_v are given in Tables V and VI, respectively. The rotational constants given in Table VII are weighted averages of values given in Tables IV, V, and VI, the weighting being proportional to the observed intensity values given in Table I. (In the work here, information from stronger bands is undoubtedly more reliable.) The uncertainties given in Table VII are the standard deviations from the weighted-average process. The uncertainties should be

TABLE III

Average Experimental Vibrational Spacings $\Delta G'$ and $\Delta G''$ (in cm^{-1}) in the $A^1\Sigma^+$ and $X^1\Sigma^+$ States of ^{7}LiH

	$\Delta G'(v+\frac{1}{2})$ in $A\,^1\Sigma^+$		$\Delta G''(v+\frac{1}{2})$ in $X\,^1\Sigma^+$	
$v+\frac{1}{2}$	Table II	Ref. 4	Table II	Ref. 4
1/2	<u>280.84</u>	280.81		<u>1359.76</u>
3/2	<u>312.97</u>	312.81		<u>1314.85</u>
5/2	<u>335.69</u>	335.45	<u>1270.87</u>	1270.91
7/2	352.74	<u>352.79</u>	<u>1227.82</u>	1228.92
9/2	365.92	<u>365.85</u>	<u>1185.40</u>	1185.33
11/2	375.72	<u>375.60</u>	<u>1143.83</u>	
13/2	382.94	<u>382.68</u>	<u>1102.63</u>	
15/2		<u>387.55</u>	<u>1061.74</u>	
17/2		<u>390.37</u>	<u>1021.16</u>	
19/2		<u>391.59</u>	<u>980.42</u>	
21/2		<u>391.05</u>	<u>939.97</u>	
23/2		<u>389.19</u>	<u>897.97</u>	
25/2		<u>385.94</u>		
27/2		<u>381.32</u>		
29/2		<u>375.21</u>		

Note. Recommended values are underlined.

TABLE IV

Rotational Constants, B_v, for ^{7}LiH

v' \ v''	2	3	4	5	6	7	8	9	10	11	12
0		2.8464	2.8501	2.8433	2.8484	2.8482	2.8493	2.8464	2.8424	2.8490	
		6.7787	6.5782	6.3784	6.1817	5.9880	5.7941	5.6005	5.4082	5.2167	
		6.7826	6.5802	6.3753	6.1813	5.9882	5.7958	5.6012	5.4093	5.2158	
1	2.8916	2.8916	2.8855	2.8879	2.8893	2.8874	2.8887			2.8871	2.8882
	6.9857	6.7842	6.5719	6.3752	6.1817	5.9915	5.7939			5.2130	5.0440
	6.9868	6.7814	6.5713	6.3744	6.1811	5.9851	5.7894			5.2110	5.0262
2	2.9040	2.9039	2.9022	2.9052			2.9058	2.9043			
	6.9849	6.7845	6.5755	6.3817			5.7968	5.6022			
	6.9850	6.7831	6.5763	6.3812			5.7964	5.6037			
3	2.9089	2.9060	2.9082	2.9137	2.9102	2.9066	2.9081				
	6.9856	6.7785	6.5650	6.3886	6.1697	5.9933	5.7913				
	6.9854	6.7803	6.5737	6.3932	6.1972	5.9884	5.7913				
4	2.9024	2.8957		2.9019	2.8990			2.9032	2.9031		
	6.9863	6.8082		6.3779	6.1802			5.5998	5.4046		
	6.9832	6.7722		6.3740	6.1732			5.6006	5.4065		
5	2.8930		2.8982	2.8957			2.9029				
			6.5855	6.3646			5.8072				
	6.9887		6.5729	6.3778			5.8075				
6			2.8771		2.8873						
			6.5709		6.1857						
			6.5742		6.1856						
7					2.8566						
					6.1821						
					6.1797						

Note. The upper entry for each band is B'_v ($A^1\Sigma^+$) obtained from Eq. (1). The middle entry is B''_v ($X^1\Sigma^+$) obtained from Eq. (2). The lower entry is B''_v ($X^1\Sigma^+$) obtained from Eq. (3) and the upper entry value for B'_v. All values are in cm^{-1}.

particularly realistic uncertainties when more than three bands contributed values to the average.

The rotational constants B_v and the vibrational energy $G(v)$ can be represented by power series in $v + \frac{1}{2}$:

$$B_v = B_e - \alpha_e(v + \tfrac{1}{2}) + \gamma_e(v + \tfrac{1}{2})^2 + \epsilon_e(v + \tfrac{1}{2})^3 + \ldots, \quad (5)$$

$$G(v) = \omega_e(v + \tfrac{1}{2}) - \omega_e x_e(v + \tfrac{1}{2})^2 + \omega_e y_e(v + \tfrac{1}{2})^3 + \omega_e z_e(v + \tfrac{1}{2})^4 + \ldots. \quad (6)$$

By fitting (B''_v, $v'' + \frac{1}{2}$) data from Table VII the rotational parameters B_e, α_e, γ_e, and ϵ_e for the ground state were obtained, and by fitting ($\Delta G(v'' + \frac{1}{2})$, $v'' + \frac{1}{2}$) data from Table III the vibrational parameters ω_e, $\omega_e x_e$, $\omega_e y_e$, and $\omega_e z_e$ for the ground state were obtained. These results are given in Table VIII. It was desired to compare results from fitting data through $v'' = 11$ with results from fitting data through $v'' = 12$ because inclusion of data from $v'' = 12$ results in irregularities somewhat outside the accuracy of the data. For example, the first differences of consecutive $\Delta G(v'' + \frac{1}{2})$ from Table III are 44.92, 43.98, 43.05, 42.42, 41.57,

TABLE V

Rotational Constants, D_v, for ^{7}LiH

v' \ v''	2	3	4	5	6	7	8	9	10	11	12
0		11.103 8.007 8.257	11.305 7.796 7.885	10.775 7.629 7.319	11.250 7.556 7.501	11.215 7.556 7.555	11.276 7.346 7.404	11.083 7.243 7.221	10.814 7.113 7.180	11.236 7.345 7.267	
1	7.964 8.263 8.299	7.974 8.332 8.165	7.476 7.370 7.293	7.726 7.473 7.435	7.838 7.576 7.555	7.766 7.725 7.372	7.857 7.359 7.101			7.702 7.108 7.018	7.538 11.172 8.918
2	6.291 8.161 8.168	6.086 8.454 8.220	6.293 7.884 8.022	6.372 7.982 8.020			6.443 7.537 7.548	6.363 7.385 7.482			
3	5.507 8.183 8.168	5.209 7.998 8.223	5.473 7.116 7.687	6.056 8.981 9.575	5.772 5.904 9.554	5.289 7.765 7.414	5.405 7.186 7.121				
4	4.863 8.298 8.115	4.606 9.308 7.696		4.711 7.505 7.247	4.435 7.353 6.789			4.885 7.147 7.267	4.873 7.002 7.124		
5	4.513 8.284		5.209 8.297 7.778	4.700 6.995 7.733			5.092 8.063 8.095				
6			4.160 7.451 7.524		5.699 8.090 8.224						
7					4.161 7.475 7.379						

Note. The upper entry for each band is D'_v ($A^1\Sigma^+$) obtained from Eq. (1). The middle entry is D''_v ($X^1\Sigma^+$) obtained from Eq. (2). The lower entry is D''_v ($X^1\Sigma^+$) obtained from Eq. (3) and the upper entry value of D'_v. All values are in units of 10^{-4} cm^{-1}.

41.20, 40.89, 40.62, 40.74, 40.45, and 42.00 cm^{-1}. The first differences of consecutive B''_v from Table VII are 0.212, 0.209, 0.204, 0.208, 0.197, 0.196, 0.193, 0.194, 0.194, 0.193, 0.194, and 0.179 cm^{-1}. Similar parameters for the $A^1\Sigma^+$ state were not determined because there was very little change to previously existing B'_v and $\Delta G(v' + \frac{1}{2})$ data (see Tables III and VII).

V. AN ADDITIONAL ANOMALY IN $A^1\Sigma^+$ LiH

As discussed in Section IV, the lower entry for each band in Table II gives the value of $J_{\max}$ ($J_{\max}$ is the maximum value of J in the subset of band data used to obtain the molecular constants). In general it was found that if band members corresponding to values of J larger than a certain value were included in the fitting of band data to Eqs. (1) and (3), then unacceptably large standard deviations resulted. The standard deviations (σ_3) in the fitting of the (0–6), (0–7), (1–6), (1–5), (2–4), and (2–5) band data to Eq. (3) are plotted as a function of $J_{\max}$ in Fig. 1. It has been found that all ($v'-v''$) curves corresponding to a given value of v' are

essentially coincident regardless of the value of v''. For example, the σ_3 versus J_{max} curve for the (0–7) band falls on the (0–6) curve, the (0–9) curve (not shown) also falls on the (0–6) curve, the (1–4) curve (not shown) and (1–5) curve fall on the (1–6) curve, etc. The quality of the band data is such that normally σ_3 would be expected to be in the neighborhood of 0.05 cm^{-1}. It is seen that σ_3 becomes unacceptably large after J_{max} values of ~18, ~24, and ~31 for the ($0-v''$), ($1-v''$), and ($2-v''$) bands, respectively. Normally, the three terms used in Eq. (4) are sufficient for representation of rotational data, so that there appears to be an anomaly associated with sufficiently excited rotational levels in the low-lying vibrational states of the $A^1\Sigma^+$ state. Confirmation that it is the A state rather than the X state which is responsible for the anomalous behavior illustrated in Fig. 1 comes from the fact that the fitting of band data to Eq. (2) yields standard deviations (σ_2) that are essentially constant (at an appropriately low value) for all J.

TABLE VI

Rotational Constants, $H_{v'}$, for ^{7}LiH

v' \ v''	2	3	4	5	6	7	8	9	10	11	12
0		65.94	67.80	58.51	68.64	66.82	68.53	64.25	59.93	67.12	
		9.33	6.17	6.75	8.10	10.00	7.78	7.41	5.89	10.54	
		14.17	6.83	-0.22	6.56	9.95	8.17	6.26	7.14	8.82	
1	25.12	25.23	16.14	21.20	23.46	22.62	24.18			20.92	9.05
	11.46	14.28	-0.40	4.43	8.65	12.32	8.22			6.33	172.44
	11.48	11.36	-2.55	3.70	8.38	7.11	4.28			5.16	84.06
2	11.19	6.75	12.35	12.94			13.90	13.03			
	9.01	21.02	10.91	14.15			10.85	10.79			
	9.03	14.22	14.46	16.10			11.52	12.55			
3	7.54	-3.27	7.37	23.44	27.03	3.32	5.33				
	9.20	9.55	-2.80	52.48	-51.16	12.82	5.00				
	8.98	16.82	6.67	71.65	95.62	6.57	3.28				
4	3.74	0.46		-0.42	-6.44			3.51	3.28		
	12.37	27.42		3.46	3.60			5.72	5.27		
	9.32	4.67		-1.66	-8.67			8.94	7.59		
5	2.61		20.41	5.45			11.47				
			16.14	-1.73			20.40				
	11.31		11.01	10.43			20.59				
6			-1.53		52.78						
			1.73		25.21						
			2.11		33.63						
7					1.74						
					5.23						
					4.15						

Note. The upper entry for each band is H'_v ($A^1\Sigma^+$) obtained from Eq. (1). The middle entry is H''_v ($X^1\Sigma^+$) obtained from Eq. (2). The lower entry is H''_v ($X^1\Sigma^+$) obtained from Eq. (3) and the upper entry value of H'_v. All values are in units of 10^{-8} cm^{-1}.

TABLE VII

Average Values of Rotational Constants for ^{7}LiH

		Averages from Tables IV, V, and VI			Previously Reported[b]		
	v	$B_v(cm^{-1})$	$D_v(10^{-4}\ cm^{-1})$	$H_v(10^{-8}\ cm^{-1})$	$B_v(cm^{-1})$	$D_v(10^{-4}\ cm^{-1})$	$H_v(10^{-8}\ cm^{-1})$
$X^1\Sigma^+$	0				7.4061	8.464	9.92
	1				7.1941	8.348	9.69
	2	6.9856 ± 0.0015	8.21 ± 0.07	10 ± 2	6.9853	8.169	9.03
	3	6.7819 ± 0.0022	8.28 ± 0.36	15 ± 6	6.7792	7.905	
	4	6.5740 ± 0.0037	7.63 ± 0.31	5 ± 7	6.5786	7.887	9.62
	5	6.3772 ± 0.0044	7.64 ± 0.39	8 ± 10	6.3744	7.561	29.2
	6	6.1812 ± 0.0032	7.53 ± 0.38	8 ± 14			
	7	5.9885 ± 0.0023	7.56 ± 0.12	10 ± 2			
	8	5.7950 ± 0.0038	7.40 ± 0.22	9 ± 4			
	9	5.6013 ± 0.0011	7.28 ± 0.10	8 ± 2			
	10	5.4082 ± 0.0014	7.13 ± 0.05	7 ± 1			
	11	5.2145 ± 0.0022	7.21 ± 0.13	8 ± 2			
	12	5.0351 ± 0.0089	10 ± 1	128 ± 44			
$A^1\Sigma^+$	0	2.8474 ± 0.0023	11.15 ± 0.17	66 ± 3	2.8500	10.590	82.0
	1	2.8883 ± 0.0018	7.76 ± 0.15	22 ± 3	2.8899	7.848	23.2
	2	2.9042 ± 0.0012	6.29 ± 0.13	11 ± 3	2.9039	6.256	11.6
	3	2.9077 ± 0.0016	5.39 ± 0.18	4 ± 7	2.9090	5.530	8.2
	4	2.9004 ± 0.0028	4.72 ± 0.16	1 ± 4	2.9055	5.030	6.0
	5	2.8966 ± 0.0036	4.80 ± 0.28	8 ± 6	2.8971	4.775	5.9
	6	2.8811 ± 0.0050	4.76 ± 0.75	20 ± 26	2.8801	4.559	6.7

Note. The averages in this table are weighted averages, the weighting being proportional to the observed intensity values given in Table I. The uncertainties are the standard deviations from the weighted-average process. Recommended values are underlined.

In order to examine the nature of this anomaly more closely, the curves shown in Fig. 2 were obtained by term analysis. The term $T_v(J) = T_e + G(v) + F_v(J)$ is defined here as the energy of a vibrational–rotational level with respect to the minimum in the $X^1\Sigma^+$ potential. Terms determined from experimental data were fit by least squares to Eq. (4), using various values of J_{max}, as shown in Fig. 2. For example, as discussed above, σ_3 becomes unacceptably large after a J_{max} value of ~18 for $(0-v'')$ bands. However, some anomalous behavior may be incipient at a J value of 16 or 17. Thus the experimental $T_0(J)$ terms of the $A^1\Sigma^+$ state were fit using $J_{max} = 15$. The magnitude of the anomalous effect at $v' = 0$ was taken to be the difference between terms extrapolated from this

TABLE VIII

Vibrational and Rotational Spectroscopic Constants Obtained from Fits to the Preceding Recommended Data Using Eqs. (5) and (6) for the $X^1\Sigma^+$ State of ^{7}LiH

constant	v" = 0 - 11 fit	v" = 0 - 12 fit
ω_e	1405.86	1406.24
$\omega_e x_e$	23.3718	23.5505
$\omega_e y_e$	0.21496	0.24174
$\omega_e z_e$	-0.0056	-0.0068
B_e[a]	7.5161	7.5134
α_e	0.21894	0.21582
γ_e	0.002719	0.001977
ε_e	-0.000094	-0.000048

[a] R_e = 1.5954 and 1.5957 Å, respectively.

Note. All values are in cm^{-1}.

fit and the experimental terms. As a possible example of what *normal* extrapolation error might be, a $v' = 2$, $J_{max} = 15$ curve is shown in Fig. 2, because the anomalous behavior for $v' = 2$ does not set in until $J \sim 31$. Thus it can be seen that the *normal* extrapolation error is very small compared to the discrepancies shown in Fig. 2 at J levels for which σ_3 is unacceptable.

There are two conceivable explanations for the anomalous behavior of the experimental term values illustrated in Fig. 2: (i) The shape of the $A^1\Sigma^+$ potential is so anomalous that Eq. (4) cannot adequately represent the term values for high J, and/or (ii) the $A^1\Sigma^+$ state is perturbed by another electronic state (e.g., $X^1\Sigma^+$).

To test the first possibility, *theoretical* term values were examined. These term values were obtained from a reasonably accurate hybrid potential curve (*8*) by numerical solution of the radial Schrödinger equation (*8*). They were then fit to Eq. (4) in exactly the same way the experimental term values were fit. These results are also shown in Fig. 2. It can be seen that corresponding theoretical and experimental curves are very similar; they are virtually identical for $v' = 2$ while for $v' = 1$ they are displaced by approximately three J units. Since the calculation of the theoretical terms assumes only the $A^1\Sigma^+$ potential and no perturbative interaction from other electronic states, it appears that the low-v', high-J anomaly under discussion here is attributable to a breakdown of the rotational expansion (Eq. (4)) for the anomalous $A^1\Sigma^+$ potential rather than an external perturbation. Thus this anomaly would seem to fall in the same category as the well-known anomalies of the $A^1\Sigma^+$ state in which $\omega_e x_e$ and α_e have negative values.

VI. GROUND-STATE POTENTIAL ENERGY CURVE

The empirical RKR turning points for the $J = 0$, $X^1\Sigma^+$ state of ^{7}LiH were calculated using a computer program reported by Tellinghuisen (*9*). The results from

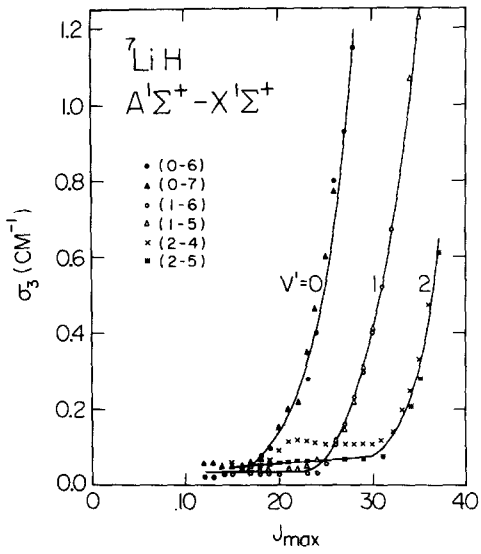


FIG. 1. The standard deviation in the fitting of band data in Eq. (3) as a function of J_{max}. Normally, σ_3 would be expected to be in the neighborhood of 0.05 cm^{-1}. Note that bands with the same value of v' yield the same function.

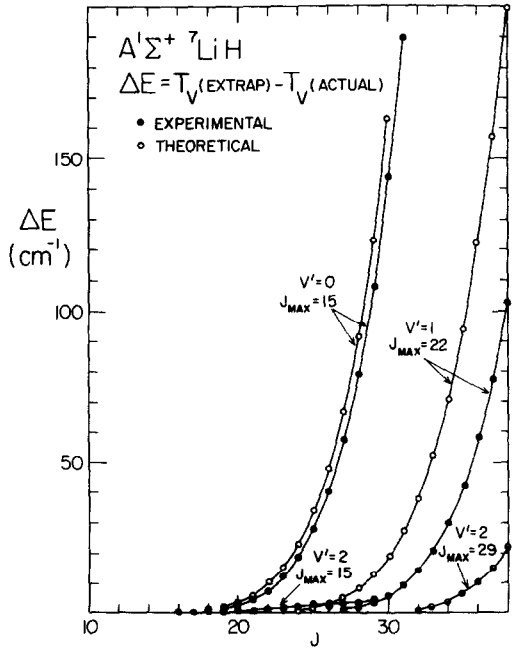


FIG. 2. Error from extrapolation of terms for the $A^1\Sigma^+$ state as a function of J. Error from extrapolation of $v' = 2$ terms through $J = 15$ is small since anomalous behavior does not set in for this vibrational level until about $J = 31$. Error from extrapolation of the other term sets shown here is anomalously large. Note that the error functions from using theoretical terms are very similar to the experimental functions (they overlap for $v' = 2$, $J_{max} = 29$ on this scale).

TABLE IX

RKR Potential Energy Curve of the $X^1\Sigma^+$ State of ^7LiH[a]

v	G(v) + Y_{00} (cm^{-1})	R_- (Å)	R_+ (Å)
0	698.12	1.4458	1.7780
1	2058.01	1.3532	1.9367
2	3372.78	1.2967	2.0607
3	4643.54	1.2548	2.1717
4	5871.27	1.2213	2.2760
5	7056.74	1.1934	2.3765
6	8200.60	1.1694	2.4748
7	9303.32	1.1486	2.5722
8	10365.19	1.1301	2.6694
9	11386.35	1.1136	2.7672
10	12366.78	1.0987	2.8662
11	13306.30	1.0851	2.9669
12	14204.55	1.0727	3.0702

[a] R_e = 1.5957 Å.

using the four rotational and four vibrational parameters in Table VIII associated with data from $v'' = 0$ through $v'' = 12$ as data input are given in Table IX (use of the parameters obtained with the data through $v'' = 11$ yielded a virtually identical potential).

The zero-point energy (zpe) associated with a given potential curve is obtainable from the spectroscopic data through the expression

$$zpe = G(0) + Y_{00}, \tag{7}$$

where, from Eq. (6),

$$G(0) = \frac{\omega_e}{2} - \frac{\omega_e x_e}{4} + \frac{\omega_e y_e}{8} + \frac{\omega_e z_e}{16} + \ldots \tag{8}$$

and Y_{00} is a Dunham correction (7)

$$Y_{00} = \frac{B_e}{4} + \frac{\alpha_e \omega_e}{12 B_e} + \frac{\alpha_e^2 \omega_e^2}{144 B_e^3} - \frac{\omega_e x_e}{4} \tag{9}$$

(note that the factor of 4 in the last term is missing in Ref. 7, p. 109). Using Eqs. (7)–(9) and data from Table VIII, the values calculated from Y_{00}, $G(0)$, and zpe, respectively, are 1.00, 697.11, and 698.11 cm^{-1} from the data through $v'' = 11$ and 0.86, 697.26, and 698.12 cm^{-1} from the data through $v'' = 12$. Analysis of Eqs. (7) and (8) clearly showed that the largest conceivable uncertainty in zpe is 0.3 cm^{-1}.

It should be noted with regard to Ref. (*4*) that the above definition of zpe is to be preferred and that Table XV of Ref. (*4*), in particular, should have been labeled "$G(0)$ values" rather than "zero point energies." Tables XVI–XXIII of Ref. (*4*) should have in column 3 the quantity "$G(v) + Y_{00}$" rather than "$G(v)$."

VII. DISCUSSION

The $X^1\Sigma^+$ state of ^{7}LiH has now been observed up to a vibrational level ($v'' = 12$) representing 70% of the height of its potential well, a fact derived from the accurately known dissociation energy for this state ($D_0 = 19588.7 \pm 0.5$ cm^{-1}) (*10*). Ultimately it is desired to observe this state right up to its dissociation limit. Since the photographed spectrum analyzed here extended to ~8700 Å, vibrational levels should have been observed up to $v'' = 14$ based on:

1. extrapolation of the extremely regular $\Delta G(v'' + \frac{1}{2})$ values (although a definite irregularity was seen in the last ΔG value corresponding to the vibrational spacing from $v'' = 11$ to 12 (see discussion near the end of Section IV));
2. a reasonable extension of intensities along the path of a primary Condon parabola (even higher v'' might have been found on secondary parabolas).

However $v'' = 13$ and 14 have not been found and the one observed band associated with $v'' = 12$ was unaccountably weak. It might be relevant to recall the derivation of the $A^1\Sigma^+$ and $X^1\Sigma^+$ states from the interaction of zero-approximation pure-atomic and pure-ionic potential curves (*11*) (also see Ref. (*7*), p. 376). The potentials for the A and X states at nuclear separations $\lesssim 3$ Å are parallel to the pure curves, while at ~3 Å the A and X states change character due to the intersection of the pure curves. Thus at nuclear separations $\gtrsim 3$ Å the A and X curves are distorted with respect to the pure curves. This distortion affects the lowest portion of the A-state potential and causes the anomalies in the A state (see Section V). However, the X-state potential would not become distorted except in its higher portion with nuclear separations $\gtrsim 3$ Å. From Table IX it is seen that for $v'' = 11$ and $v'' = 12$, R_+ values are 2.97 and 3.07 Å, respectively.

APPENDIX: ASSIGNED LINES OF THE $A^1\Sigma^+$–$X^1\Sigma^+$ BANDS OF ^{7}LiH

The lines given here have been assigned to bands of the $A^1\Sigma^+$–$X^1\Sigma^+$ electronic transition of ^{7}LiH. The values of all R and P lines are given as vacuum wavenumbers in reciprocal centimeters. The numbers in parentheses at the head of each pair of branches define the band in terms of v' and v'', respectively. The superscripts indicate which investigators identified lines of the band according to the following: (a) F. H. Crawford and T. Jorgensen, Jr., (b) K. C. Li and W. C. Stwalley, (c) F. B. Orth and W. C. Stwalley.

The lines given here corresponding to a and b are taken from the Ph.D. theses of T. Jorgensen (Harvard University, 1935) and K. C. Li (University of Iowa, 1976) and tabulated in K. C. Li and W. C. Stwalley, "Assigned Lines of the $A^1\Sigma^+$–$X^1\Sigma^+$ Bands of ^{6}LiH, ^{6}LiD, ^{7}LiH, and ^{7}LiD," United States Energy Research and Development Administration Report C00-2326-20 (1976). (Note that some of the blends indicated in the report appear to be erroneous, but are given correctly here.) Abbreviations: *, blend; OBSD, obscured.

(2- 0)[a]

J	R(J)	P(J)
0	26542.81	
1	26481.87	26522.20
2	26527.35	26498.19
3	26505.99	26465.37
4		26424.01
5	26436.46	26373.43
6	26368.13	26314.01
7	26330.78	26245.82
8	26264.51*	26168.82
9	26189.90*	26083.15
10	26105.29	25988.85
11	26012.14	25885.97
12	25857.47	25774.62
13		25654.79

(3- 0)[a]

J	R(J)	P(J)
0	26878.61	
1	26875.43	26857.86
2	26863.20	26834.18
3	26842.00	26801.53
4	26811.74	26759.81
5	26772.46	26709.30
6	26724.36	26649.99
7	26667.25	26581.77
8	26601.26	26504.88
9	26526.28	26419.60
10	26442.75*	26325.57
11	26350.32	26223.15*
12	26249.27	26112.18
13	26139.52	25992.98
14	26021.48	

(4- 0)[a]

J	R(J)	P(J)
0	27232.04*	
1	27228.19	27210.77
2	27215.98	27186.97
3	27194.72	27154.21
4		27112.56
5	27125.26	27062.08
6	27077.08	27002.75
7	27020.07	26934.62
8	26954.14	26857.86
9	26879.42	26772.45*
10	26795.93	26678.53
11		26576.04
12	26603.03	
13	26493.55	26346.39
14	26375.97	26219.30
15	26249.96	26084.19
16	26115.68	25941.09
17	25973.24	

(5- 0)[a]

J	R(J)	P(J)
0	27600.80	
1	27597.20	27576.61
2	27581.67	27552.79
3	27560.40	27520.01
4	27530.06	27478.30
5	27490.76	27427.71
6	27442.54*	27368.29
7	27385.38	27300.08
8	27319.36	27223.23
9	27244.50	27137.71
10	27161.02	27043.71
11	27068.80	26941.26
12	26968.08	26830.49
13	26858.83	26711.46
14	26742.48	26584.24

(6- 0)[a]

J	R(J)	P(J)
0	27972.81	
1		27952.23
2	27957.14	27928.39
3	27935.59	27895.55
4	27905.24	27853.75
5	27865.76	27803.00
6	27817.59	27743.41
7	27759.93	27675.07
8	27693.74	27598.01
9	27618.55	27512.33
10	27534.96	27418.19
11	27442.58*	
12	27339.17	27204.44
13	27232.04	27085.17
14	27114.29	26957.89
15	26988.27	
16		26679.47
17	26711.92	26528.53

(7- 0)[a]

J	R(J)	P(J)
0	28355.59	
1	28352.07	28335.72
2	28340.38	28317.02
3	28317.98	28278.11
4	28287.34	28236.20
5	28247.64	28185.33
6	28198.91	28125.54
7	28141.30	28056.96
8	28074.75	27979.67
9	27999.40	27893.65
10	27915.22	27799.06
11	27822.29	27696.05
12	27721.03	27584.67
13	27611.17	27465.07
14	27492.56	27337.36
15	27366.47	27201.63
16	27231.37	27058.11
17	27089.22	26906.80
18	26938.78	26748.00
19		26581.77*

(8- 0)[a]

J	R(J)	P(J)
0	28742.87	
1	28739.47	28722.46
2	28726.83	28698.53
3	28705.03	28665.51
4	28674.71	28623.40
5	28634.12	28572.25*
6	28585.11	28512.29
7	28527.00	28443.43
8	28460.07	28365.79
9	28384.22	28279.42
10	28299.62	28184.42
11	28206.29	28080.94
12	28104.30	27969.02
13	27993.95	27848.92
14		27720.67
15	27747.87*	27584.42
16	27612.65	27440.15
17	27469.38	
18		27128.77
19	27159.38	

(9- 0)[a]

J	R(J)	P(J)
0	29133.24	
1	29129.61	29112.86
2	29116.45*	29088.78
3	29094.73	29055.73
4	29063.59	29013.44
5	29023.29	28962.20
6	28973.94	28901.86
7	28915.36	28832.59
8	28847.35	
9		28667.71
10	28686.18	28572.25
11	28591.34	28468.16
12	28489.53	28355.58*
13	28378.35	28234.86
14	28258.74	28105.91
15	28130.35	27969.02*
16	27994.79	27823.84
17	27850.53	27671.15
18	27698.45	27510.79
19	27538.67	27343.02
20	27371.16	

(10- 0)[a]

J	R(J)	P(J)
0	29524.75	
1	29521.02	29504.44*
2	29508.02	29480.35
3	29485.78	29447.09
4	29454.18	29404.60
5	29413.47	29352.98
6	29363.56	29292.39
7	29304.43	29222.72
8	29236.24	29144.14
9	29159.24	29056.74
10	29073.27	28960.72
11	28978.52	28855.93
12	28876.19	28742.87*
13	28762.97	28621.20
14	28642.39	
15	28513.47	28353.05
16	28376.34	28207.69
17	28231.06	28053.84
18	28077.47	27892.44
19	27916.94	27723.53
20	27747.87*	

(11- 0)[a]

J	R(J)	P(J)
0	29915.76	
1	29911.86	29895.48
2	29898.61	29871.34
3	29876.00	29837.94
4	29844.06	29795.26
5	29802.85	29743.30
6	29752.40	29682.29
7	29692.77	29612.20
8	29624.04	29533.21
9	29546.20	29445.12
10	29459.39	29348.26
11	29363.56*	29242.95
12	29259.30	29128.63
13	29146.15	29006.38
14	29024.52	28874.99
15	28894.44	
16	28756.02	

(12- 0)[a]

J	R(J)	P(J)
0	30304.85	
1	30300.90	
2	30287.26	30260.44
3		30226.85
4	30231.99	30183.89
5	30190.27	30131.63
6	30139.21	30070.19
7	30078.90	29999.62
8	30009.39	29919.93
9	29930.72	29831.27
10	29843.00	29733.75
11	29746.35	29627.45
12	29640.87	29512.47
13	29526.59	29389.04
14	29403.71	29257.16
15	29272.33	29116.95*
16	29132.54	28968.81
17	28984.58	28812.63
18	28828.45	

(13- 0)[a]

J	R(J)	P(J)
0	30640.73	
1	30686.31	30670.60
2	30672.66	30646.27
3	30649.35	30612.48
4	30616.54	30569.25
5	30574.26	30516.69
6	30522.57	30454.75
7	30461.50	30384.07
8	30391.14	30303.30
9	30311.58	30213.83
10	30222.83	
11	30125.02	30008.21
12	30018.36	29892.27
13	29902.82	29767.74
14	29778.56	29634.71
15	29645.75	29493.27
16	29504.44*	29343.67
17	29354.81	29186.02
18	29197.03	29020.56
19	29031.23	28847.28
20	28858.50*	28666.54
21	28676.27	

(14- 0)[a]

J	R(J)	P(J)
0		
1	31067.48	31052.09
2	31053.35	31027.57*
3	31029.66	30993.52
4	30996.37	30950.04
5	30953.54	30896.99
6	30901.11	30834.63
7	30839.26	30762.80
8	30767.97	30681.80
9	30687.37	30591.55
10	30597.50	30492.27
11	30498.53	30383.57
12	30390.50	
13		30141.18
14	30147.74	30006.78
15	30013.30	29864.44
16	29870.28	29712.90
17	29718.87	29553.64
18	29559.87	29386.43
19	29391.34	

(15- 0)[a]

J	R(J)	P(J)
0		
1		
2		
3		
4		
5		
6		
7	31210.79	
8	31138.48	
9	31056.72	30963.09
10	30965.67	30862.84
11	30865.39	30753.47
12	30755.90	30635.22
13	30637.39	30508.04
14	30509.92	30372.22
15	30373.65	30227.64
16	30228.69	30074.57
17	30075.05	

(1- 1)[a]

J	R(J)	P(J)
0		
1	24867.23	
2	24855.53	
3	24835.59	24795.41
4	24806.75	24755.34
5	24769.23	24706.73
6	24723.06	24649.68
7	24668.32	24583.32
8	24604.54	24510.08*
9	24532.30*	24427.64*
10	24451.31	24336.88
11	24361.63	24237.78
12	24263.40	24130.32
13	24156.52	24014.71
14	24041.06	23890.87
15	23917.15	23758.91
16	23784.79	23618.90
17	23644.16	

(2- 1)[a]

J	R(J)	P(J)
0	25183.05	
1	25180.42	25162.95
2	25168.93	25139.66
3	25148.91	25108.45
4	25120.29	25068.51
5	25083.10	25020.05
6	25037.31	24963.15
7	24982.90	24897.89
8	24919.97	24824.25
9	24848.51	24742.29
10	24768.56	24652.27
11	24680.15	24553.97
12	24584.06	24447.61
13	24478.20	24333.23
14	24364.73	24210.85
15	24243.08	24080.61
16	24113.28	23942.61
17		23796.88

(3- 1)[a]

J	R(J)	P(J)
0	25519.84	
1	25516.35	25498.60
2	25504.71	25475.76
3	25484.78	25444.24
4	25456.28	25405.08
5	25419.10	25355.88
6	25373.48	25299.07
7	25319.30	25233.94
8	25256.71	
9	25185.64	25038.80
10	25106.12	24989.01
11	25018.33	24931.07
12	24922.26	24865.16
13	24817.97	24791.37
14	24705.57	24709.74
15	24585.10	24620.10
16	24456.70	24523.44*
17	24320.39	24419.93
18	24176.36	23987.02
19	24024.61	23827.83
20	23865.41	23661.93
21	23698.92	23490.12

(4- 1)[a]

J	R(J)	P(J)
0		
1	25868.84	
2	25857.07	25828.46
3	25837.52	25796.97
4	25809.48	25757.08
5	25771.91	25708.63
6	25726.25	25651.86
7	25672.17	25586.71
8	25609.24	25513.29*
9	25538.56	25431.63
10	25459.36	25341.91
11	25371.77	25244.13
12	25276.01	25138.38
13	25172.11	25024.80
14		24903.49
15	24947.30	24774.52
16	24812.57	24638.00
17	24670.05	24494.11
18	24531.35	24342.40
19	24383.32	
20	24226.33	24019.14

J	R(J)	(0- 5)$^{b,c}_{P}$(J)
1		
2	19588.75	19571.61
3	19588.25	19552.09
4	19564.65	19525.18
5	19541.79	19481.25
6	19511.68	19450.39
7	19478.99	19415.27
8	19439.28	19376.99
9	19397.27	19331.65
10	19351.77	19287.10
11	19302.53	19240.32
12	19251.07	19190.76
13	OBSD	18906.45
14	18847.61	18863.42
15	18790.61	18805.14
16	18732.14	18747.73
17	18672.59	18725.59
18	18446.29*	18691.93
19	18385.35	18629.35
20	18029.89	17800.66
21	OBSD	17603.36
22	OBSD	17520.38
23	17557.50	17192.67
24		
25		

J	R(J)	(1- 5)$^{b,c}_{P}$(J)
1		
2	19868.89	19852.80*
3	19861.69	19806.24
4	19846.41	19802.96
5	19824.16	19772.93
6	19794.32	19734.83
7	19758.48	19690.61
8	19717.68	19630.45
9	19664.02	19569.56
10	19606.25	19501.65
11	19540.39	19435.00*
12	19467.33	19361.11
13	19390.22	19284.08
14	19303.82*	19162.08
15	19210.69	19060.45
16	19109.26*	18957.65
17	18999.15	18845.52
18	18886.33	18731.99*
19	18768.17	18588.17
20	18640.96	18433.99*
21	18386.76	18167.41
22	18220.31	18015.16
23	18067.89	17837.19
24	17900.00*	17654.69
25	17746.00*	17524.69
26	17576.93	17350.70
27	17402.74	17177.78
28	17224.97	17000.13
29	17042.63	16808.13
30	16845.18	16608.19
31	OBSD	16412.18
32	16465.42*	16009.64
33	16264.60	15801.82
34	16054.73	15595.16
35	15852.40	15394.01
36	15643.30*	15171.34
37	15231.34	

J	R(J)	(2- 5)$^{b,c}_{P}$(J)
1		
2	20159.73	20085.78
3	20137.70	
4		
5	OBSD	19998.32
6	20027.39	19944.36
7	19979.43	19883.60
8	19922.39	19811.64
9	19857.36	19734.49
10	19781.16	19647.29
11	19710.16	19557.49
12	19625.76	19480.73
13	19534.42	19380.56
14	19431.64	19276.04
15	19320.36	19161.85
16	19202.61	19044.85
17	19078.35	18915.63
18	18947.11	18782.98
19	18809.93	18643.85
20	18664.83	18504.85
21	18513.63	18356.02
22	18355.72	18201.73
23	18191.23	18041.72
24	18019.99	17864.27
25	17840.99	17700.35
26	17654.32	17503.31*
27	17460.04	17315.58*
28	17258.32	17117.37
29	17049.03	16907.21
30	16831.99	16678.03
31	17042.86	16591.11*
32	16850.98	16389.29
33	16656.57	16180.29
34	16456.73	15983.98
35	16250.73	15775.02*
36	16040.76	15565.31
37	15824.40*	15353.66
38	15604.30*	
39	15423.40	

J	R(J)	(3- 5)$^{c}_{P}$(J)
1		
2	20872.70	20854.22
3	OBSD	20834.51
4	20848.58	20774.61*
5	20826.38	20647.46
6	20781.34	20663.11
7	20718.86	20572.69
8	OBSD	20505.06
9	20614.72	20435.81
10	20479.37	20351.77
11	20402.87	20265.14
12	20319.53	20172.88
13	20233.58	20084.82
14	20143.58	19996.79
15	20030.98	19856.42
16	19922.39*	19739.06
17		19607.83
18		19497.09
19		

J	R(J)	(4- 5)$^{c}_{P}$(J)
1		
2	21213.95	
3	21191.90	
4		
5	21126.66	21052.48
6	21081.87	20999.66
7	21026.77	20933.11
8	20977.94*	20870.94*
9	20917.88	20807.98
10	20844.51	20711.98
11	20767.90	OBSD
12	20684.82	20537.37
13	20599.84	20452.86
14	20499.84	20352.86
15	20396.65	OBSD
16	20277.93	20104.45
17	20162.47	19987.83
18	20025.81	19718.69
19	19893.37	19578.08
20	19765.25	19434.05
21	19636.57	
22	19362.57	
23		
24		

J	R(J)	(0- 6)$^{c}_{P}$(J)
1		
2	18437.63	18409.29
3	OBSD	OBSD
4	18401.82	18364.50
5	18371.64	18312.80
6	18329.39	18274.82
7	18283.23	18215.64
8	18241.01	18169.92
9	18191.01	18119.48
10	18142.20	18067.24
11	18097.68	18015.24
12	18052.38	17952.48
13	17999.13	17899.22
14	17947.45	17847.89*
15	17899.45	17744.89*
16	17847.25	17722.90*
17	17787.25	17690.05*
18	17741.81*	17606.05*
19		16685.10
20		
21		
22	16830.94*	16635.98
23	16681.87	16481.36
24	16524.40	16326.66
25	OBSD	16256.62
26	16203.30	15986.95
27	16034.06	15812.79
28	15883.42	15654.30
29	OBSD	15485.74*
30	OBSD	15265.74*
31	15316.06	15076.12*
32		
33		
34		

J	R(J)	(1- 6)$^{c}_{P}$(J)
1		
2	18705.02	18632.86
3	18694.26	18594.32
4	18652.84	18554.34
5	18625.42	18497.84
6	18567.82	18447.80
7	18534.31	18399.80
8	18479.39	18335.43
9	18431.04*	18284.28
10	18376.73	18243.65
11	18195.49	18053.71
12	18107.62	17957.43
13	17912.02	17756.00*
14	17804.38	17611.12
15	17694.04	17500.14
16	17647.24	17453.27
17	17772.80	17393.32
18	17311.13	17111.81*
19		
20		
21	17028.88	16818.18
22	16874.93	16663.68
23	16724.93	16503.68
24	16565.29	16339.11
25	16400.94	16169.92
26	16228.00	15987.40
27	16048.00	15818.24
28	15861.82	15637.84
29	15701.04	15422.29
30	15519.86	15202.50*
31	15329.42	
32	15139.42	
33		
34		

J	R(J)	(2- 6)$^{c}_{P}$(J)
1		
2		
3		
4		
5		
6		
7		
8		
9		
10		
11		
12		
13		
14		
15		
16	18240.40	17938.68
17	18135.57	17837.84
18	18024.91	17594.41*
19	17944.35	17549.29
20	17855.51	17449.29
21	17780.17	17308.54
22	17720.44	17162.78
23	17644.44	17011.48*
24	17084.44	
25	16928.65*	
26		

J	R(J)	(3- 6)$^{c}_{P}$(J)
1		
2	19367.94	
3	19354.33	
4	19333.78	
5	19306.63	
6		
7	19233.14	19198.76
8	19191.35	19147.58
9	19136.32	19090.14
10	19073.98	19026.59
11	19013.35	18960.14
12	18950.67	18880.67
13	18880.67	18798.53
14		
15		
16		
17		
18		
19		
20		
21		
22		
23		
24		

J	R(J)	(4- 6)$^{c}_{P}$(J)
1		
2	19720.70	
3	19707.11*	19666.41
4	19688.52	OBSD
5	19652.74*	19556.33
6	19625.26*	19525.36
7	19585.89	19500.45
8	19548.50	19443.12
9	19539.51	19379.46
10	19486.45	19333.72
11	19430.34	19251.66
12	19389.34	OBSD
13	19321.15	19170.03
14	19316.27	19097.26
15	19239.68	18976.63
16	18939.68	18765.23
17	OBSD	18654.35
18	18729.02	18537.84
19	18695.05	18488.49
20	18370.02	18156.72
21		

J	R(J)	(5- 6)$^{c}_{P}$(J)
1		
2		
3		
4		
5		
6		
7		
8		
9		
10	19726.35	
11	19654.38	
12	19576.40	
13	19492.07	
14		
15		
16	19305.40	

J	R(J)	$P_b^C(J)$
0	20461.97	
1	20447.72*	
2	20394.85	20375.68
3	20349.81	20329.19
4	OBSD	20293.19
5	20356.74	20240.82
6	20228.87	20118.33
7	20228.60	20118.06
8	OBSD	20044.15
9	20100.00	19972.93
10	20028.07	19890.64
11		19802.59
12		19700.58

J	R(J)	$P_b^C(J)$
0	17383.13	17326.05
1	17336.24	17309.98
2	OBSD	17283.39
3	17303.12	17252.63
4	17274.06	17215.60*
5	17224.84	17172.60
6	17204.64	17122.84
7	17168.57	17066.96
8	17105.29	17004.71
9	17105.29	17004.71
10	16980.06	16881.44*
11	16907.22	16779.88
12	16827.54	16662.19
13	16827.54	16662.19
14	16648.33	16497.33
15	16648.33	16497.33
16	16548.91	OBSD
17	16443.17	16278.83
18	16333.16	16160.28
19	16088.97	15905.83
20	16088.97	15905.83
21	15959.23	15769.99
22	15883.99	15688.07
23	15883.99	15688.07
24	15732.62	15531.92
25	15387.07	15175.98
26	15072.50*	15015.65

J	R(J)	$P_b^C(J)$
0		
1		
2		
3	20830.18*	20816.09*
4	20809.48*	OBSD
5	20784.97*	20758.25
6	20784.97*	20674.62
7	20707.19	20623.06
8	20659.97	OBSD
9	20606.42	20500.61
10	20606.42	20500.61
11	20488.02	20353.66
12	20407.46	20270.97
13	20328.62	20182.42
14	20328.62	20182.42
15	20152.68	19987.90
16	20055.93	19892.13
17	19953.26	19770.85
18	19834.95	19654.28
19	19730.95	19405.67
20		

J	R(J)	$P_b^C(J)$
0		
1		
2		
3	17617.72	
4	17604.92*	
5	17585.51	
6		
7	OBSD	17406.18
8	17495.15*	OBSD
9	17446.59	17290.22
10	17394.83	17226.27
11	17337.65	17161.10
12	17304.24*	17071.80
13	17212.91	16986.08
14	17094.37	16895.24
15	OBSD	16695.46
16	16885.34*	16587.06
17	16760.28	16472.81
18	16653.33	16352.93
19	16538.80	16224.27
20	16538.80	16071.10
21	16298.11	16096.78*
22	16165.93	15960.53
23	16040.91	15813.67
24	15884.23	15644.24
25	15744.23	15523.03
26	15594.01	15367.84
27	15439.81*	15208.95
28	15117.83	15044.95

J	R(J)	$P_b^C(J)$
0		
1		
2		
3		17901.47
4		OBSD
5	17899.10	17847.26
6	OBSD	17810.97
7	17855.47	17768.08*
8	17761.14	17664.82*
9	17711.03	17598.63
10		17536.53
11		17486.53
12		17389.58
13		17304.58

J	R(J)	$P_b^C(J)$
0		
1		
2		
3		
4		
5		
6		
7		

$^cQ(J)$ (0- 9)

J	R(J)	Q(J)
0		
1		
2	15255.67	15227.35
3	15244.60	15205.15
4	15227.93	15177.37*
5	15205.48	15144.33
6	15177.37*	15105.67
7	15143.27	15061.53
8	15103.33	15011.68
9	15057.44	OBSD
10	15005.54	OBSD
11	OBSD	14828.88
12	OBSD	14756.56
13	14818.15	14678.82
14	14739.29	14595.50
15	14656.89	14506.44
16	14569.52	14412.03
17	14476.57	14312.26
18	14377.91	
19	14274.13	14097.00
20	14165.07	13981.94
21	14050.74	13861.58
22		13736.82
23		13607.68
24	13680.09	13474.18
25	13547.89	13336.79
26	13411.83	13195.57
27	13272.13	13050.87
28	13129.03	12902.92
29	12982.86	12751.95
30	12833.84	12598.34
31	12682.35	12442.37
32	12528.70	

$^cQ(J)$ (2- 9)

J	R(J)	Q(J)
0		
1		
2	15850.24	15821.29
3	OBSD	OBSD
4	OBSD	15772.13
5	15802.51	15739.54*
6	OBSD	15701.59
7	15743.50	15658.50
8	15795.81	15610.13
9	15662.57	15556.61
10	15514.13	15497.91
11	15560.28	15434.13
12	15501.01*	15365.25
13	15436.34	15291.20*
14	15366.16	15212.42
15	15291.20*	15128.62
16	15210.60	15039.84*
17	15124.83	
18	15034.02	

$^cQ(J)$ (4- 9)

J	R(J)	Q(J)
0		
1		
2	16538.91	
3	OBSD	
4	16512.53	16460.67
5	16491.30	16428.14*
6	16464.66	OBSD
7	OBSD	OBSD
8	OBSD	OBSD
9	OBSD	16245.88
10	16305.07	16187.60
11	16251.95	16124.23
12	OBSD	16056.04
13	OBSD	15982.94
14	OBSD	15905.13
15	15988.32	15822.61
16	15909.93	15735.33
17	15826.60*	15643.41
18	15738.52	15547.46

$^cQ(J)$ (0-10)

J	R(J)	Q(J)
0		
1		
2		
3		
4		
5		
6		
7		
8		
9		
10		
11		
12		
13	13868.70	13733.20
14		13655.13
15	13722.33	13571.88
16	13647.06	13483.50
17	13554.48	13390.15
18	13462.77	13291.91
19	13366.03	13188.83
20	13264.49	13081.18
21	13158.19	12968.95
22	13047.45	12852.58
23	12932.52	12732.04
24	12813.53	12607.65
25	12690.79	12479.65
26	12564.52	12348.00
27	12434.96	12213.87

$^cQ(J)$ (2-10)

J	R(J)	Q(J)
0		
1		
2		
3		
4		
5		
6		
7		
8		
9		
10		
11		
12		
13		
14		
15		
16		
17		
18		
19		
20		13737.57*
21		13633.18
22	13737.57*	13525.17
23	13631.34	13413.27
24	13521.24	13297.63
25	OBSD	13178.59
26	13290.02	13056.24

$^cQ(J)$ (4-10)

J	R(J)	Q(J)
0		
1		
2		
3		
4		
5		
6	15492.76*	
7	15463.91	
8	15428.70	15332.39*
9	15389.60	15282.69
10	15345.57	15227.93*
11	15296.71	15168.99
12	15243.09	
13	15184.67	
14	15121.45	OBSD
15	15053.60	OBSD
16	OBSD	
17	OBSD	14721.38
18	14823.26	14632.18

$^cQ(J)$ (0-11)

J	R(J)	Q(J)
0		
1		
2	13337.76	
3	13328.90*	
4	13315.47	
5	13296.98	13235.72
6	13273.44*	13201.70
7	13244.75	13162.96
8	13211.00*	13119.41
9	13172.07	13071.00
10	13127.95	13017.74
11	13078.61	12959.58*
12		12896.72
13	12964.23	12828.89
14	12899.26	12756.21
15	12829.21	12678.80
16	12754.16	12596.61
17	12674.18	12509.83
18	12589.41	12418.50
19	12500.06	
20	12406.14	

$^cQ(J)$ (1-11)

J	P(J)	Q(J)
0		
1		
2		13530.40
3	13610.90	13570.76
4	13597.93	13546.47
5	13580.20*	13517.64
6	13557.58	13484.09*
7	13530.30	OBSD
8	13498.10	13403.54
9	13461.10	13356.51
10	13419.24	13304.85
11	13372.59	13248.68
12	13321.16*	13188.03
13	OBSD	13122.91
14	13203.44	13053.26
15	13137.46	12979.26
16	13066.69	12900.83
17	12991.29	12818.03
18	12911.31	12731.07
19	12826.82	12639.93
20	12738.06	12544.73

$^cQ(J)$ (1-12)

J	R(J)	Q(J)
0		
1		
2		
3		
4	12703.69	12652.34
5	12688.03	
6	12667.77	12594.14
7	12643.20*	12559.09
8	12614.27	12519.78
9	12580.83	12476.29
10	12542.93	12428.58
11	12500.67	

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